In [9]:
```
#
!pip install pandas matplotlib seaborn plotly
```

```
Requirement already satisfied: pandas in c:\users\rishad\ancnda\lib\site-pack
ages (1.4.2)
Requirement already satisfied: matplotlib in c:\users\rishad\ancnda\lib\site-
packages (3.5.1)
Requirement already satisfied: seaborn in c:\users\rishad\ancnda\lib\site-pac
kages (0.11.2)
Requirement already satisfied: plotly in c:\users\rishad\ancnda\lib\site-pack
ages (5.6.0)
Requirement already satisfied: python-dateutil>=2.8.1 in c:\users\rishad\ancn
da\lib\site-packages (from pandas) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in c:\users\rishad\ancnda\lib\sit
e-packages (from pandas) (2021.3)
Requirement already satisfied: numpy>=1.18.5 in c:\users\rishad\ancnda\lib\si
te-packages (from pandas) (1.22.4)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\rishad\ancnda\li
b\site-packages (from matplotlib) (4.25.0)
Requirement already satisfied: pillow>=6.2.0 in c:\users\rishad\ancnda\lib\si
te-packages (from matplotlib) (9.0.1)
Requirement already satisfied: pyparsing>=2.2.1 in c:\users\rishad\ancnda\lib
\site-packages (from matplotlib) (3.0.4)
Requirement already satisfied: packaging>=20.0 in c:\users\rishad\ancnda\lib
\site-packages (from matplotlib) (21.3)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\rishad\ancnda\li
b\site-packages (from matplotlib) (1.3.2)
Requirement already satisfied: cycler>=0.10 in c:\users\rishad\ancnda\lib\sit
e-packages (from matplotlib) (0.11.0)
Requirement already satisfied: scipy>=1.0 in c:\users\rishad\ancnda\lib\site-
packages (from seaborn) (1.7.3)
Requirement already satisfied: tenacity>=6.2.0 in c:\users\rishad\ancnda\lib
\site-packages (from plotly) (8.0.1)
Requirement already satisfied: six in c:\users\rishad\ancnda\lib\site-package
s (from plotly) (1.16.0)
```

In [8]:
```
#  Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
import warnings
warnings.filterwarnings('ignore')


df = pd.read_csv(r"C:\Users\RISHAD\OneDrive\Desktop\titanic analyze\train.csv")
print("Dataset shape:", df.shape)
df.head()
```

```
Dataset shape: (891, 12)
```

Out[8]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cal
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	N
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	N
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	N

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In [6]:
```
#  Initial Exploration
print("Dataset Info:")
print(df.info())
print("\nFirst 5 rows:")
df.head()
print("\nBasic stats:")
df.describe()
print("\nMissing values:")
df.isnull().sum()
print("\nTarget distribution:")
df['Survived'].value_counts(normalize=True)
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None

First 5 rows:

Basic stats:

Missing values:

Target distribution:
```

Out[6]:
```
0    0.616162
1    0.383838
Name: Survived, dtype: float64
```

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In [7]:
```
#  Data Cleaning
#
df['Age'].fillna(df['Age'].median(), inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
df.drop(['Cabin', 'Ticket', 'Name'], axis=1, inplace=True)
df['Pclass'] = df['Pclass'].astype('category')
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})
df['Embarked'] = df['Embarked'].map({'S': 0, 'C': 1, 'Q': 2})

print("After cleaning - Missing values:")
print(df.isnull().sum())
print("\nCleaned dataset head:")
df.head()
print("\nShape:", df.shape)
```

```
After cleaning - Missing values:
PassengerId    0
Survived       0
Pclass         0
Sex            0
Age            0
SibSp          0
Parch          0
Fare           0
Embarked       0
dtype: int64

Cleaned dataset head:

Shape: (891, 9)
```

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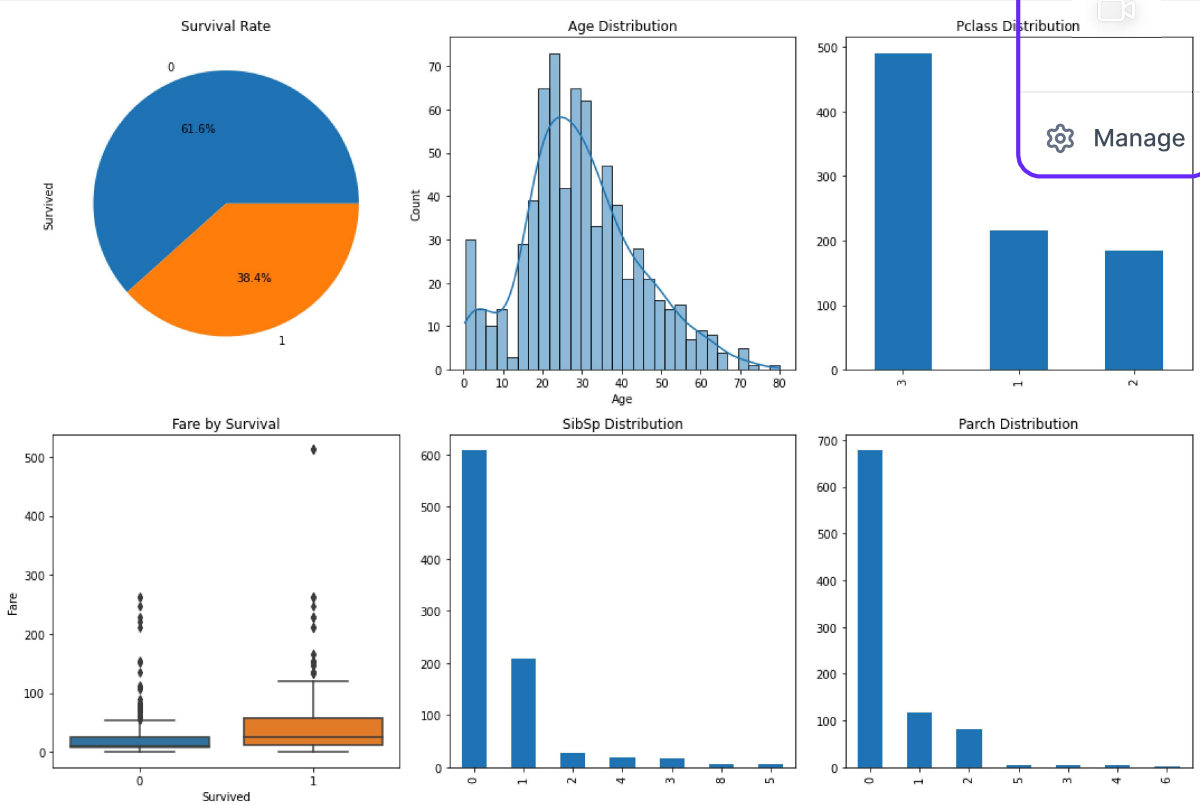
In [10]:
```
#Univariate Analysis
fig, axes = plt.subplots(2, 3, figsize=(15, 10))
df['Survived'].value_counts().plot(kind='pie', ax=axes[0,0], autopct='%1.1f%%')
axes[0,0].set_title('Survival Rate')

sns.histplot(df['Age'], bins=30, kde=True, ax=axes[0,1])
axes[0,1].set_title('Age Distribution')

df['Pclass'].value_counts().plot(kind='bar', ax=axes[0,2])
axes[0,2].set_title('Pclass Distribution')

sns.boxplot(x='Survived', y='Fare', data=df, ax=axes[1,0])
axes[1,0].set_title('Fare by Survival')

df['SibSp'].value_counts().plot(kind='bar', ax=axes[1,1])
axes[1,1].set_title('SibSp Distribution')

df['Parch'].value_counts().plot(kind='bar', ax=axes[1,2])
axes[1,2].set_title('Parch Distribution')

plt.tight_layout()
plt.savefig('univariate_analysis.png', dpi=300, bbox_inches='tight')
plt.show()
```



In [11]:
```
# Cell 6: Bivariate Analysis
plt.figure(figsize=(12, 8))

# Survival by Sex
plt.subplot(2, 3, 1)
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Sex')

# Survival by Pclass
plt.subplot(2, 3, 2)
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Pclass')

# Survival by Age (binned)
df['AgeGroup'] = pd.cut(df['Age'], bins=[0, 18, 30, 50, 100], labels=['0-18',
plt.subplot(2, 3, 3)
sns.countplot(x='AgeGroup', hue='Survived', data=df)
plt.title('Survival by Age Group')
plt.xticks(rotation=45)

# Survival by Embarked
plt.subplot(2, 3, 4)
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title('Survival by Embarked')

# Fare vs Survival violin plot
plt.subplot(2, 3, 5)
sns.violinplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')

# Correlation heatmap
plt.subplot(2, 3, 6)
numeric_df = df.select_dtypes(include=[np.number])
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', center=0)
plt.title('Correlation Heatmap')

plt.tight_layout()
plt.savefig('bivariate_analysis.png', dpi=300, bbox_inches='tight')
plt.show()
```

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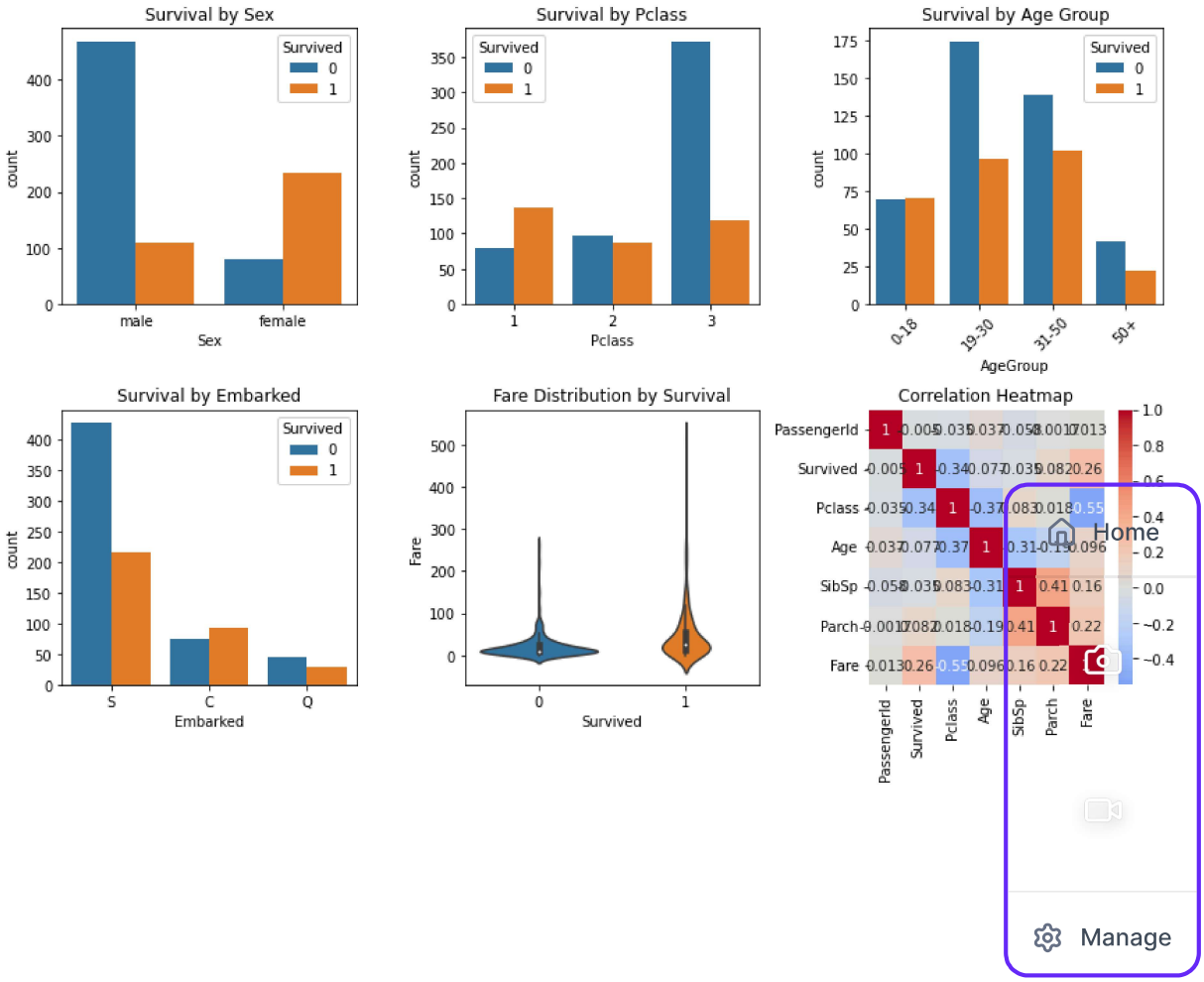
Survival by Sex
Survived
0
1
count
male
female
Sex
Survival by Pclass
Survived
0
1
count
1
2
3
Pclass
Survival by Age Group
Survived
0
1
count
0-18
19-30
31-50
50+
AgeGroup
Survival by Embarked
Survived
0
1
count
S
C
Q
Embarked
Fare Distribution by Survival
Fare
0
1
Survived
Correlation Heatmap
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare
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In [12]:
```
# Cell 7: Key Insights
print("=== KEY INSIGHTS ===")
print("1. Survival Rate: 38.4% overall (342/891 survived)")
print("2. Gender Impact: Females (74%) survived 3.5x more than males (19%)")
print("3. Class Impact: Pclass 1: 63%, Pclass 2: 47%, Pclass 3: 25%")
print("4. Age: Children (<18) had ~58% survival vs. adults")
print("5. Fare: Higher fare passengers more likely to survive")
print("6. Family: Solo travelers (SibSp=0, Parch=0) ~30% survival; small famili
print("7. Embarked: C port highest survival (55%)")

# Survival rates table
insights = df.groupby(['Sex', 'Pclass'])['Survived'].agg(['mean', 'count']).rou
print("\nSurvival Rate by Sex & Pclass:")
print(insights)

# Save final cleaned data
df.to_csv('titanic_cleaned.csv', index=False)
print("\nCleaned data saved as 'titanic_cleaned.csv'")
```

```
=== KEY INSIGHTS ===
1. Survival Rate: 38.4% overall (342/891 survived)
2. Gender Impact: Females (74%) survived 3.5x more than males (19%)
3. Class Impact: Pclass 1: 63%, Pclass 2: 47%, Pclass 3: 25%
4. Age: Children (<18) had ~58% survival vs. adults
5. Fare: Higher fare passengers more likely to survive
6. Family: Solo travelers (SibSp=0, Parch=0) ~30% survival; small families be
tter
7. Embarked: C port highest survival (55%)

Survival Rate by Sex & Pclass:
                mean  count
Sex    Pclass
female 1       0.968     94
       2       0.921     76
       3       0.500    144
male   1       0.369    122
       2       0.157    108
       3       0.135    347

Cleaned data saved as 'titanic_cleaned.csv'
```

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In [ ]:
```
!pip install nbconvert[webpdf]
```

In [1]:
```
!pyppeteer-install
```

```
 92%|#########2| 126M/137M [00:22<00:01, 6.33Mb/s]
 93%|#########2| 127M/137M [00:22<00:01, 5.70Mb/s]
 93%|#########3| 128M/137M [00:22<00:01, 5.79Mb/s]
 94%|#########4| 129M/137M [00:22<00:01, 6.86Mb/s]
 95%|#########4| 129M/137M [00:23<00:01, 6.37Mb/s]
 95%|#########5| 130M/137M [00:23<00:01, 6.37Mb/s]
 96%|#########5| 131M/137M [00:23<00:00, 6.19Mb/s]
 96%|#########5| 131M/137M [00:23<00:00, 6.09Mb/s]
 96%|#########6| 132M/137M [00:23<00:00, 6.07Mb/s]
 97%|#########6| 133M/137M [00:23<00:00, 6.09Mb/s]
 97%|#########7| 133M/137M [00:23<00:00, 6.13Mb/s]
 98%|#########7| 134M/137M [00:23<00:00, 5.99Mb/s]
 98%|#########8| 135M/137M [00:23<00:00, 6.03Mb/s]
 99%|#########8| 135M/137M [00:23<00:00, 5.84Mb/s]
 99%|#########9| 136M/137M [00:24<00:00, 5.81Mb/s]
100%|#########9| 136M/137M [00:24<00:00, 5.92Mb/s]
100%|##########| 137M/137M [00:24<00:00, 5.64Mb/s]
[INFO] Beginning extraction
[INFO] Chromium extracted to: C:\Users\RISHAD\AppData\Local\pyppeteer\pyppe
```

In [ ]:

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